1. Extract Sample document and apply following document preprocessing methods: Tokenization, POS Tagging, stop words removal, Stemming and Lemmatization.
2. Create representation of documents by calculating Term Frequency and Inverse DocumentFrequency.

In [1]:
```
# Import libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.feature_extraction.text import TfidfVectorizer
import re

pd.options.display.max_columns = None
sns.set(style='whitegrid', palette='muted')
```

In [2]:
```
# Download necessary NLTK data
import nltk
import ssl

try:
    _create_unverified_https_context = ssl._create_unverified_context
except AttributeError:
    pass
else:
    ssl._create_default_https_context = _create_unverified_https_context

# Download NLTK data
nltk.download('punkt', quiet=True)
nltk.download('punkt_tab', quiet=True)
nltk.download('stopwords', quiet=True)
nltk.download('wordnet', quiet=True)
nltk.download('wordnet_ic', quiet=True)
nltk.download('averaged_perceptron_tagger', quiet=True)
nltk.download('averaged_perceptron_tagger_eng', quiet=True)
nltk.download('omw-1.4', quiet=True)

from nltk.tokenize import word_tokenize, sent_tokenize
from nltk.corpus import stopwords
from nltk.stem import PorterStemmer, WordNetLemmatizer
from nltk import pos_tag

print('NLTK data ready.')
```

```
NLTK data ready.
```

In [3]:
```
# Create sample documents
sample_documents = [
    "Natural language processing is a subfield of linguistics, computer scienc
```

```
    "Machine learning is the study of computer algorithms that improve automat
    "Data science combines statistics, programming, and domain expertise to ex
    "Artificial intelligence aims to create machines that can perform tasks th
    "Text mining is the process of deriving high-quality information from text
    "Sentiment analysis determines the emotional tone behind a series of words
    "Named entity recognition identifies and classifies named entities in text
    "Topic modeling discovers abstract topics in a collection of documents.",
    "Word embeddings represent words as vectors in a continuous vector space."
    "Chatbots use natural language processing to simulate human conversation."
]

# Create DataFrame
df = pd.DataFrame({'document': sample_documents})
df.head()
```

Out[3]:

	document
0	Natural language processing is a subfield of l...
1	Machine learning is the study of computer algo...
2	Data science combines statistics, programming,...
3	Artificial intelligence aims to create machine...
4	Text mining is the process of deriving high-qu...

In [4]:

```
# EDA:
print('Shape:', df.shape)
print('Size:', df.size)
```

```
Shape: (10, 1)
Size: 10
```

In [5]:

```
print('\nInfo:')
df.info()
```

```
Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 1 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   document  10 non-null     object
dtypes: object(1)
memory usage: 208.0+ bytes
```

In [6]:

```
print('\nDescribe (text lengths):')
df['text_length'] = df['document'].apply(len)
df['text_length'].describe()
```

```
Describe (text lengths):
```

Out[6]:
```
count     10.000000
mean      91.900000
std       32.219559
min       70.000000
25%       72.500000
50%       74.000000
75%       98.750000
max      173.000000
Name: text_length, dtype: float64
```

In [7]:
```python
print('\nSample documents:')
for i, doc in enumerate(df['document'][:3]):
    print(f'Doc {i+1}: {doc[:100]}...')
```

```
Sample documents:
Doc 1: Natural language processing is a subfield of linguistics, computer scien
ce, and artificial intellige...
Doc 2: Machine learning is the study of computer algorithms that improve automa
tically through experience....
Doc 3: Data science combines statistics, programming, and domain expertise to e
xtract insights from data....
```

In [8]:
```python
# Insert rows with null/empty values in between the dataset
null_rows = pd.DataFrame([
    {'document': 'This is a sample document with some content.'},
    {'document': ''},
    {'document': np.nan}
])

insert_position = 3
df_with_null = pd.concat([df.iloc[:insert_position], null_rows, df.iloc[insert

print('New shape with inserted rows:', df_with_null.shape)
df_with_null.head(10)
```

```
New shape with inserted rows: (13, 2)
```

Out[8]:

	document	text_length
0	Natural language processing is a subfield of l...	173.0
1	Machine learning is the study of computer algo...	99.0
2	Data science combines statistics, programming,...	98.0
3	This is a sample document with some content.	NaN
4		NaN
5	NaN	NaN
6	Artificial intelligence aims to create machine...	113.0
7	Text mining is the process of deriving high-qu...	74.0
8	Sentiment analysis determines the emotional to...	74.0
9	Named entity recognition identifies and classi...	74.0

In [9]:

```
print('\nNull counts before removal:')
print(df_with_null.isnull().sum())
print('\nEmpty documents:', (df_with_null['document'] == '').sum())
```

```
Null counts before removal:
document       1
text_length    3
dtype: int64

Empty documents: 1
```

In [10]:

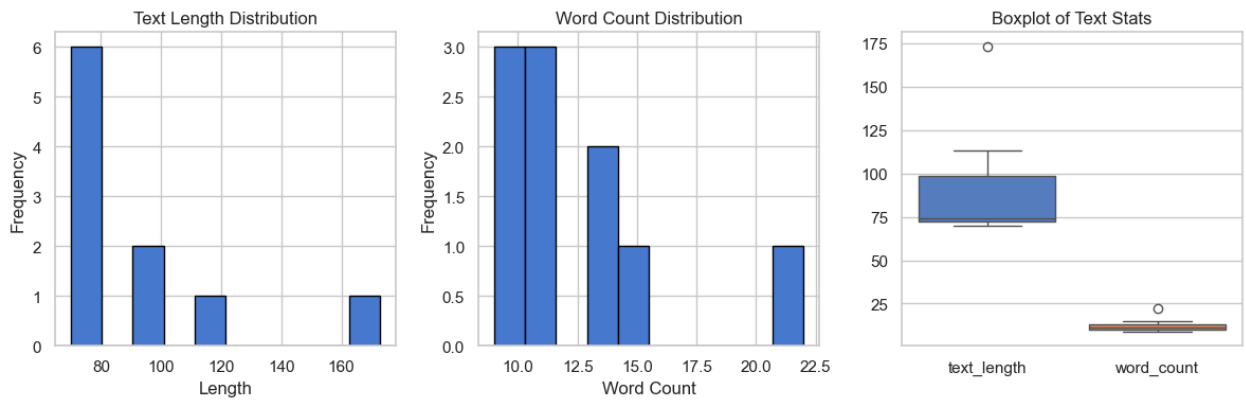
```
# Remove null/empty values and confirm
df_no_null = df_with_null.dropna().copy()
df_no_null = df_no_null[df_no_null['document'] != ''].reset_index(drop=True)
print('Shape after cleanup:', df_no_null.shape)
print('\nNull counts after cleanup:\n', df_no_null.isnull().sum(), sep="")
print('\nEmpty documents after cleanup:', (df_no_null['document'] == '').sum()
```

```
Shape after cleanup: (10, 2)

Null counts after cleanup:
document       0
text_length    0
dtype: int64

Empty documents after cleanup: 0
```

In [11]:

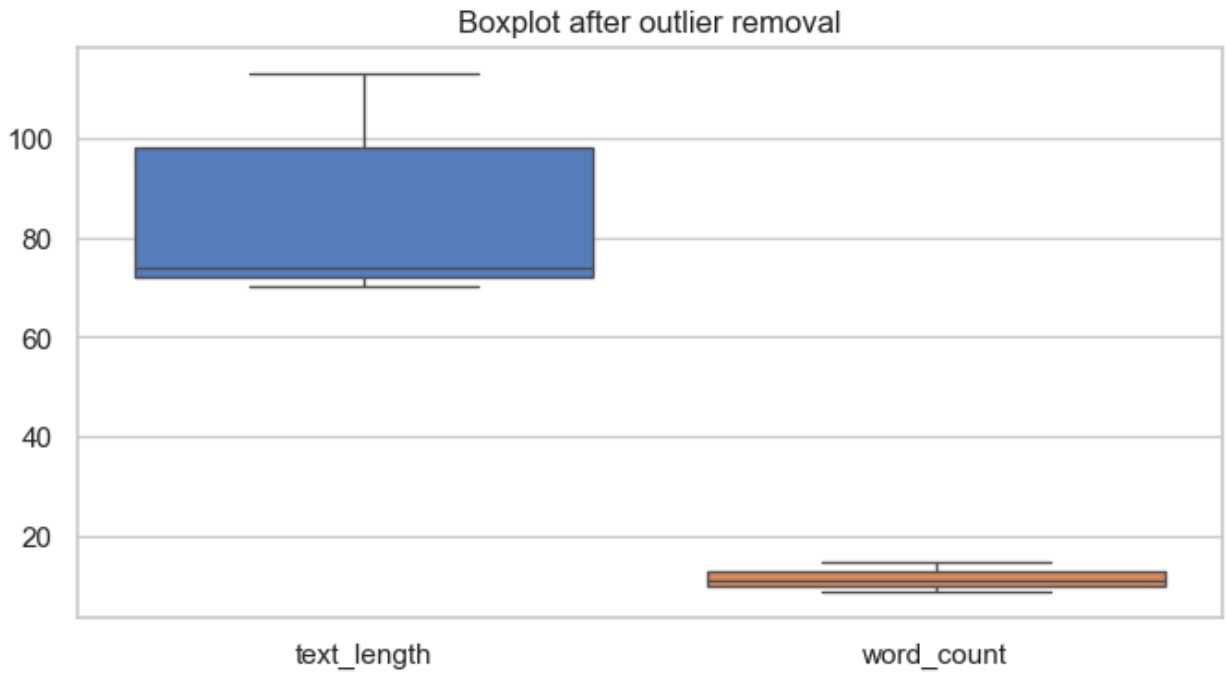
```
# Visualizations: text length distribution, word count distribution
df_no_null['word_count'] = df_no_null['document'].apply(lambda x: len(x.split(
df_no_null['text_length'] = df_no_null['document'].apply(len)

plt.figure(figsize=(12, 4))

plt.subplot(1, 3, 1)
```

```
plt.hist(df_no_null['text_length'], bins=10, edgecolor='black')
plt.title('Text Length Distribution')
plt.xlabel('Length')
plt.ylabel('Frequency')

plt.subplot(1, 3, 2)
plt.hist(df_no_null['word_count'], bins=10, edgecolor='black')
plt.title('Word Count Distribution')
plt.xlabel('Word Count')
plt.ylabel('Frequency')

plt.subplot(1, 3, 3)
sns.boxplot(data=df_no_null[['text_length', 'word_count']])
plt.title('Boxplot of Text Stats')

plt.tight_layout()
plt.show()
```



In [12]:
```
# Outlier detection and removal (based on text length)
df_clean = df_no_null.copy()

# Remove outliers using IQR on text_length
q1 = df_clean['text_length'].quantile(0.25)
q3 = df_clean['text_length'].quantile(0.75)
iqr = q3 - q1
lower = q1 - 1.5 * iqr
upper = q3 + 1.5 * iqr
df_clean = df_clean[(df_clean['text_length'] >= lower) & (df_clean['text_lengt

print('Shape after outlier cleanup:', df_clean.shape)

plt.figure(figsize=(8,4))
sns.boxplot(data=df_clean[['text_length', 'word_count']])
plt.title('Boxplot after outlier removal')
plt.show()
```

```
Shape after outlier cleanup: (9, 3)
```



In [13]:
```
# Category checks (no categories in this dataset, but we can add document type
df_clean['doc_type'] = ['technical'] * 5 + ['general'] * 4
print('Document types:')
print(df_clean['doc_type'].value_counts())
```

```
Document types:
doc_type
technical    5
general      4
Name: count, dtype: int64
```

In [14]:
```
df_clean.head()
```

Out[14]:

	document	text_length	word_count	doc_type
1	Machine learning is the study of computer algo...	99	13	technical
2	Data science combines statistics, programming,...	98	13	technical
3	Artificial intelligence aims to create machine...	113	15	technical
4	Text mining is the process of deriving high-qu...	74	11	technical
5	Sentiment analysis determines the emotional to...	74	11	technical

In [15]:
```
# Convert to numeric
from sklearn.preprocessing import LabelEncoder
le = LabelEncoder()
df_clean['doc_type_num'] = le.fit_transform(df_clean['doc_type'])
```

```
print('Type mapping:', dict(zip(le.classes_, le.transform(le.classes_))))
```

```
Type mapping: {'general': np.int64(0), 'technical': np.int64(1)}
```

In [16]:
```
# Preprocessing: Clean text (lowercase, remove punctuation)
def clean_text(text):
    text = text.lower()
    text = re.sub(r'[^\w\s]', '', text)
    return text

df_clean['cleaned_text'] = df_clean['document'].apply(clean_text)
df_clean.head()
```

Out[16]:

	document	text_length	word_count	doc_type	doc_type_num	cleaned_te
1	Machine learning is the study of computer algo...	99	13	technical	1	machi learning is t study comput algc
2	Data science combines statistics, programming,...	98	13	technical	1	data scien combin statisti programmi a
3	Artificial intelligence aims to create machine...	113	15	technical	1	artific intelligen aims crea machine
4	Text mining is the process of deriving high-qu...	74	11	technical	1	text mining the process derivi highqua
5	Sentiment analysis determines the emotional to...	74	11	technical	1	sentime analys determin the emotior tc

Task 1: Document preprocessing methods

In [17]:
```
# Select first document for demonstration
sample_doc = df_clean['cleaned_text'].iloc[0]
print('Original sample document:')
print(sample_doc)
```

```
Original sample document:
machine learning is the study of computer algorithms that improve automatically
through experience
```

In [18]:
```
# Tokenization
tokens = word_tokenize(sample_doc)
```

```
print('Tokenization:')
print(tokens)
```

```
Tokenization:
['machine', 'learning', 'is', 'the', 'study', 'of', 'computer', 'algorithms',
'that', 'improve', 'automatically', 'through', 'experience']
```

In [19]:
```
# POS Tagging
pos_tags = pos_tag(tokens)
print('POS Tagging:')
print(pos_tags)
```

```
POS Tagging:
[('machine', 'NN'), ('learning', 'NN'), ('is', 'VBZ'), ('the', 'DT'), ('study',
'NN'), ('of', 'IN'), ('computer', 'NN'), ('algorithms', 'NN'), ('that', 'WDT'),
('improve', 'VBP'), ('automatically', 'RB'), ('through', 'IN'), ('experience',
'NN')]
```

In [20]:
```
# Stop words removal
stop_words = set(stopwords.words('english'))
filtered_tokens = [word for word in tokens if word not in stop_words]
print('After stop words removal:')
print(filtered_tokens)
```

```
After stop words removal:
['machine', 'learning', 'study', 'computer', 'algorithms', 'improve', 'automati
cally', 'experience']
```

In [21]:
```
# Stemming
stemmer = PorterStemmer()
stemmed_tokens = [stemmer.stem(word) for word in filtered_tokens]
print('Stemming:')
print(stemmed_tokens)
```

```
Stemming:
['machin', 'learn', 'studi', 'comput', 'algorithm', 'improv', 'automat', 'exper
i']
```

In [22]:
```
# Lemmatization
lemmatizer = WordNetLemmatizer()
lemmatized_tokens = [lemmatizer.lemmatize(word) for word in filtered_tokens]
print('Lemmatization:')
print(lemmatized_tokens)
```

```
Lemmatization:
['machine', 'learning', 'study', 'computer', 'algorithm', 'improve', 'automatic
ally', 'experience']
```

In [23]:
```
# Task 2: TF-IDF representation
documents = df_clean['cleaned_text'].tolist()

# Create TF-IDF vectorizer
tfidf_vectorizer = TfidfVectorizer(stop_words='english', max_features=100)
tfidf_matrix = tfidf_vectorizer.fit_transform(documents)

# Get feature names (terms)
```

```
feature_names = tfidf_vectorizer.get_feature_names_out()

# Convert to DataFrame for better visualization
tfidf_df = pd.DataFrame(tfidf_matrix.toarray(), columns=feature_names)
print('TF-IDF Matrix shape:', tfidf_matrix.shape)
print('Top 10 terms:', feature_names[:10])
tfidf_df
```

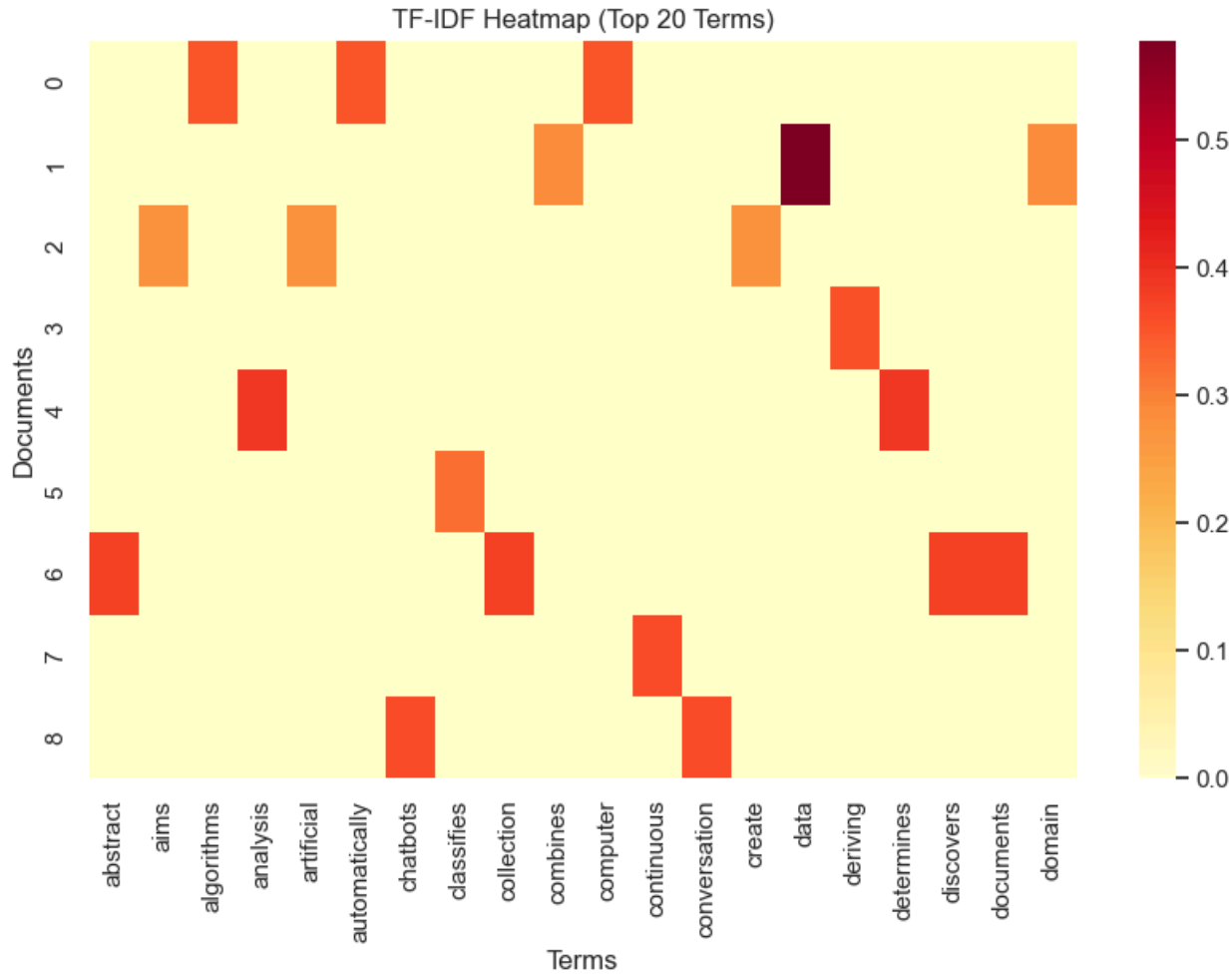
```
TF-IDF Matrix shape: (9, 67)
Top 10 terms: ['abstract' 'aims' 'algorithms' 'analysis' 'artificial' 'automati
cally'
 'chatbots' 'classifies' 'collection' 'combines']
```

Out[23]:

	abstract	aims	algorithms	analysis	artificial	automatically	chatbots
0	0.000000	0.000000	0.353553	0.000000	0.000000	0.353553	0.000000
1	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
2	0.000000	0.280459	0.000000	0.000000	0.280459	0.000000	0.000000
3	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
4	0.000000	0.000000	0.000000	0.385949	0.000000	0.000000	0.000000
5	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
6	0.377964	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
7	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
8	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.360062

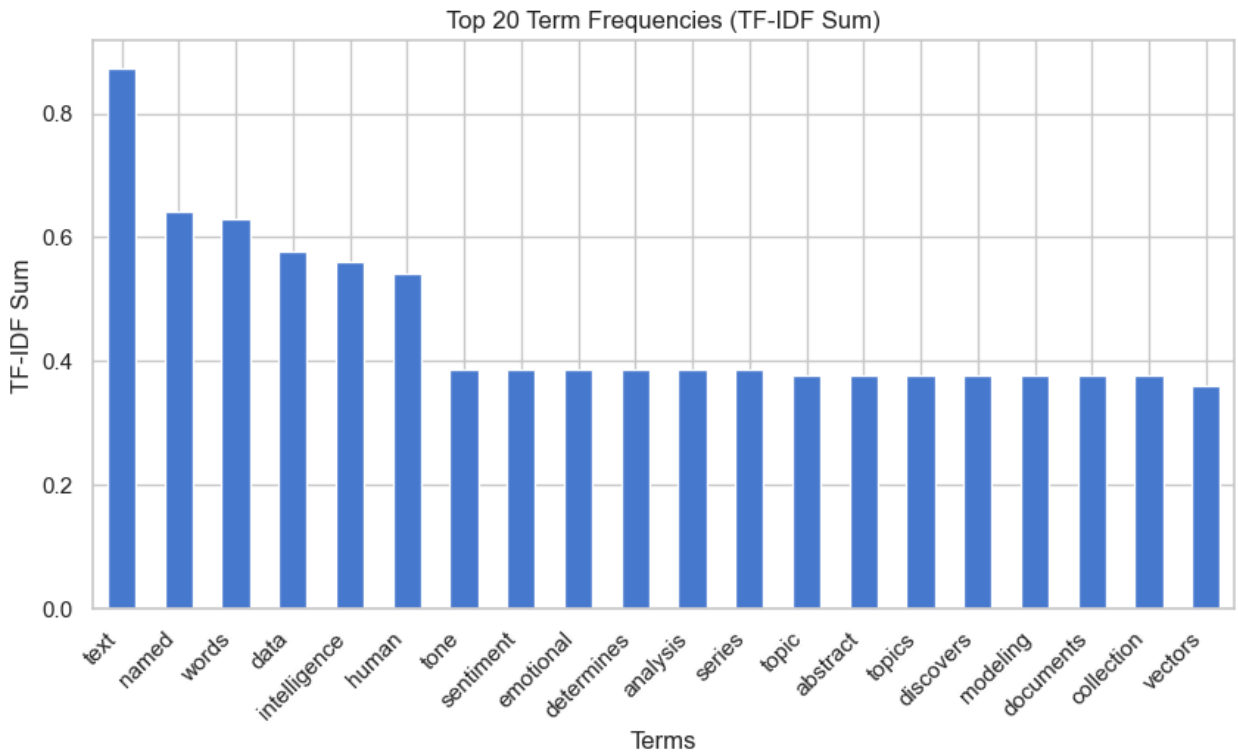
In [24]:

```
# Visual insights: TF-IDF heatmap for top terms
top_terms = feature_names[:20]
tfidf_top = tfidf_df[top_terms]

plt.figure(figsize=(10, 6))
sns.heatmap(tfidf_top, annot=False, cmap='YlOrRd', cbar=True)
plt.title('TF-IDF Heatmap (Top 20 Terms)')
plt.xlabel('Terms')
plt.ylabel('Documents')
plt.show()
```



In [25]:

```
# Term frequency analysis
term_frequencies = tfidf_df.sum(axis=0).sort_values(ascending=False)
plt.figure(figsize=(10, 5))
term_frequencies.head(20).plot(kind='bar')
plt.title('Top 20 Term Frequencies (TF-IDF Sum)')
plt.xlabel('Terms')
plt.ylabel('TF-IDF Sum')
plt.xticks(rotation=45, ha='right')
plt.show()
```


Top 20 Term Frequencies (TF-IDF Sum)
TF-IDF Sum
0.0
0.2
0.4
0.6
0.8
text
named
words
data
intelligence
human
tone
sentiment
emotional
determines
analysis
series
topic
abstract
topics
discovers
modeling
documents
collection
vectors
Terms